

Find and Fix: Degree–Radian and Quadrant–Sign Errors

Directions. Every problem below shows work a student actually handed in. In each one:

- first produce the *correct* answer yourself, in the form the question asks for;
- then say what went wrong, in a sentence or two. Name the step, not just the wrong number.
- Exact values (fractions and radicals) are expected unless a decimal is requested.

1. A student is asked to convert $\frac{5\pi}{6}$ radians to degrees, and writes

$$\frac{5\pi}{6} \cdot \frac{\pi}{180} = 0.0457.$$

- (a) Give the correct measure.

Answer: _____ degrees

- (b) Name the error in one sentence.

Answer: _____

2. A student evaluates $\cos 210^\circ$ this way: “the reference angle is 30° , and $\cos 30^\circ = \frac{\sqrt{3}}{2}$, so $\cos 210^\circ = \frac{\sqrt{3}}{2}$.”

(a) Give the correct exact value.

Answer: _____

(b) Explain which step is wrong and why.

Answer: _____

3. A student solves $2 \sin \theta - 1 = 0$ on the interval $0^\circ \leq \theta < 360^\circ$ and reports $\theta = 0.5236$ and $\theta = 2.6180$.

(a) Give the correct solutions.

Answer: _____ degrees

(b) Name the error in one sentence.

Answer: _____

4. A student solves $\tan \theta = -1$ on the interval $0^\circ \leq \theta < 360^\circ$. They press $\tan^{-1}(-1)$, read -45 , and report that $\theta = -45^\circ$ is the only solution.

(a) Give every solution in the interval.

Answer: _____ degrees

(b) Two separate things went wrong here. Name both.

Answer: _____

5. A student evaluates $\sin \frac{7\pi}{6}$ and writes $-\frac{\sqrt{3}}{2}$, reasoning that “the reference angle is $\frac{\pi}{6}$, and the third quadrant makes sine negative.”

(a) Give the correct exact value.

Answer: _____

(b) The sign reasoning is correct. Explain what is not.

Answer: _____

6. A student needs $\sin(2)$, where the 2 is an angle in radians. Their calculator returns 0.0349, and they write that down.

- (a) Give the correct value of $\sin(2)$ to four decimal places.

Answer: _____

- (b) Explain what the calculator actually computed, and why the two results are so far apart.

Answer: _____

7. A student is told that $\sin \theta = -\frac{3}{5}$ and that $\cos \theta > 0$. They write: “sine is negative, so θ is in the third quadrant, and there cosine is negative too, so $\cos \theta = -\frac{4}{5}$.”

- (a) Give the correct exact value of $\cos \theta$.

Answer: _____

- (b) State the correct quadrant, and explain what the student overlooked.

Answer: _____

8. A student solves $\cos \theta = -\frac{1}{2}$ on the interval $0 \leq \theta < 2\pi$ and reports $\theta = 120$ and $\theta = 240$.

(a) Give the exact solutions in radians.

Answer: _____

(b) Name both errors, and state the check that would have caught them.

Answer: _____